

CO₂ 倍增对长白赤松幼苗根-土界面水分 运输过程的影响

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摘要: 用具有非破坏性的电导率方法测定土壤水分的廓线, 与挖掘法(或打孔法)获取的根系分布对比, 研究 CO₂ 倍增条件下一年生的长白赤松(*Pinus sylvestris* Linn. var. *sylvestrisformis* (Takenouchi) Cheng et C. D. Chu) 幼苗根-土界面的水分运输状态。结果表明: (1) 土壤水分廓线由植物的活性所调制, 根系分布密集的土层其水分含量也高。(2) CO₂ 倍增, 根系-土壤水分运输的活跃层及根系分布都将向土壤深处位移。研究证明, 电导率方法能够指示发生于根-土界面上的水分运输状态, 方法简单, 且对土壤无破坏。

关键词: CO₂ 倍增; 长白赤松幼苗; 根-土界面; 水分运输; 土壤电导率

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Influence of CO₂ Doubling on Water Transport Process at Root/Soil Interface of *Pinus sylvestris* var. *sylvestrisformis* Seedlings

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Abstract: Water transport at the root/soil interface of 1 year old *Pinus sylvestris* Linn. var. *sylvestrisformis* (Takenouchi) Cheng et C. D. Chu seedlings under CO₂ doubling was studied by measuring soil electric conductance to survey soil water profiles and comparing it with root distribution surveyed by soil coring and root harvesting in Changbai Mountain in 1999. The results were: (1) The profiles of soil water content were adjusted by root activity. The water content of the soil layer with abundant roots was higher. (2) When CO₂ concentration was doubled, water transport was more active at the root/soil interface and the roots were distributed into deeper layer. It was shown in this work that the method of measuring electric conductance is an inexpensive, non-destructive and relatively sensitive way for underground water transport process.

Key words: CO₂ doubling; *Pinus sylvestris* var. *sylvestrisformis* seedlings; root/soil interface; water transport; electric conductance of soil

The transport of water across the root/soil interface may be of crucial importance for determining water availability to plant^[1]. Lack of quantitative information about the water transport in the interface has been due, in part, to the difficulty of determining moisture gradients in the immediate vicinity of roots^[2].

Elevated atmospheric CO₂ concentration affect many underground processes which may lead to higher water use efficiency (WUE)^[3], more carbon allocation to the roots and higher spatial heterogeneity in the architecture of the roots system^[4]. These underground processes are related to water movement across the root/soil interface. Therefore, water movement process will be a remarkable driving factor on other processes^[5].

If the profile of the soil moisture is measured quantitatively, we can understand the underground processes under elevated CO₂^[6]. However, it is difficult to take an accurate measurement of soil moisture profile in the field using simple techniques, such as soil coring and root harvesting. These simple methods would disturb the water state between soil and root during the measurement and could not acquire detailed information about water transport near roots. Therefore, an inexpensive, non-invasive and relatively sensitive method to monitor soil moisture profile has been desired in the field measurement^[7,8]. In this paper, we described a way of measurement by monitoring soil electrical conductance (SEC) to reveal soil water profile.

1 Materials and Methods

1.1 Plant materials and site

Seeds of *Pinus sylvestris* Linn. var. *sylvestrisformis* (Takenouchi) Cheng et C. D. Chu were sown in the experimental field at the Open Research Station of Changbai Mountain Forest Ecosystem, The Chinese Academy of Sciences. Density of one year old seedlings was kept 1 plant per 5 cm × 5 cm.

Changbai Mountain is an important ecological research site for its various natural ecosystems and rich gene pool. It has played an important role in the International Long-Term Ecological Research Network since 1993. The experimental site was located to the east of the research station (128°06'E, 42°24'N), at 738.1 m elevation on the north slope of Changbai Mountain. The soil is dark brown soil developed from volcano ash, the topography is basaltic mesa, and the parent rock is loose volcano ash sand^[9]. The vertical structure of soil is litter layer (0–3 cm), mixed layer (3–24 cm) dark gray sod, gray brown soil with volcano ash and gravel sod (gray soil), mixed layer (below 24 cm) volcano ash with gravel sod and stone (sand soil). There are natural gray soil/sand soil interfaces. In this research, one year old seedlings were selected as samples, their roots can penetrate to these interfaces. During the experiment, the litter on soil surface was removed.

1.2 CO₂ treatments

Since June 1999, rectangular open-top chambers (OTCs) (1.2 m long, 0.9 m wide and 0.9 m high) were established on the site. The basic experimental design is listed in Table 1. Daily regulation was applied to keep the CO₂ concentration constant. Each plot was watered daily at regular intervals.

Table 1 Experimental design

Treatment No.	CO ₂ treatment	Seedling arrangement
1	Ambient	With seedling
2	700 $\mu\text{mol}/\text{mol}$	With seedling
3	700 $\mu\text{mol}/\text{mol}$	No seedling
4	Ambient	No seedling

1.3 Methods

1.3.1 Soil water profile by measuring soil electrical conductance The sampling system comprised of a plexiglass club, platinum wires and electric conductance meter. The plexiglass club (1 cm in diameter) made with a series of holes (0.5 mm in diameter) was used. The distance between two holes was 3 cm. Each hole was fixed with a section of platinum wire (0.15 mm diameter and 0.5 cm long) with one end of the platinum wire 1 mm out of the hole and reached soil and the other linked to a wire that was connected to electric conductance meter. The

expositive end of each platinum wire was employed as electrode to monitor soil water. There were 10 electrodes in this system.

This system was buried vertically into the soil. The first electrode was put 2 cm deep in the mineral soil. All measurement ends of wires were laid outside of the open top chamber. Close attention was paid to avoid disturbing the soil against electrodes when buried the plexiglass club.

The seeds of *Pinus sylvestris* var. *sylvestrisformis* were sown directly in the top soil of the system, then the position of the uppermost electrode was the zero reference for the seedling roots growing down. In our research, the distribution of seedlings was horizontally homogeneous.

Water existed continuously in soil, and the soil electric conductance would change with the soil water content^[10]. The greater the electric conductance was, the higher the water content in the soil layer and vice versa. The water state at the root/soil interface might be determined by comparing the electrical conductance in that soil layer of the same thickness.

1.3.2 Roots biomass sampling Soil coring method was employed to collect root biomass samples as described by Sternberg *et al.*^[8]. Soil samples were collected using a steel tube of 5 cm in diameter at a depth interval of about 5 cm to 30 cm in depth. Root biomass was estimated from a means of three random soil coring. In this work, roots of no more than 1 mm in diameter were defined as fine-root.

Lateral root samples were collected by root harvesting method. Total length of lateral roots in a depth interval of about 3 cm was counted as the sum of all lateral roots in that soil layer. Five seedlings were randomly selected to count the statistic distribution of the total lateral root length.

1.3.3 Soil temperature profile The vertical pattern of soil temperature of 0–30 cm layer was measured using the mean of values from three digital thermometers inserted at each site. The digital thermometer was calibrated against mercury thermometer to ± 0.1 °C and the temperature sample gradient was the same as the conductance.

2 Results and Discussion

2.1 Interface characteristic of water transport

Fig. 1 shows the results of water transport and different CO₂ treatments. In treatment 4, there was almost no change in electric conductance from 0–9 cm, but increased sharply in the layer from 9 cm to 24 cm, indicating that soil water content in 0–9 cm layer remained constant and there was a trend for water movement upward in 9–24 cm layer. This pattern of soil water distribution

was completely dependent upon soil surface evaporation, because there was no plant in this treatment, the solar energy would be the only factor driving water transport.

Compared with treatment 4, SEC profile of treatment 3 became more precipitous and total water content became less under doubled CO₂ condition. Obviously, the water in 9–24 cm soil layer in treatment 4 had participated in the surface evaporation process under CO₂ stress. This result was similar to what Vose *et al.*^[11] observed that there was a noticeable soil moisture decrease by 11% under doubled CO₂ in the chamber.

The water state represented by SEC profiles in treatment 4 had been improved by the activity of roots compared with that of treatment 1. It was the seedlings that made the soil water more active in 3–18 cm soil layer and at the interface area between the gray soil and the sand soil. If soil water state was regulated by root activity, it must be correlated to the biological characteristics of seedlings (See 2.2).

Comparing treatment 2 with treatment 1, it can be inferred that the water active layer was shifted from 9 cm down to 12 cm by CO₂ stress, and the water content near the interface was higher. As stated above, the elevated CO₂ would accelerate soil water loss, however, the profile showed that water increased in the peak zone. The only explanation must be attributed to the activity of roots which changed the water status.

It was worth pointing out that the effect of soil temperature on the electric conductance could be neglected in our research because the temperature variation within every 3 cm soil layer was no more than 4 °C in the growing season, the conductivity shift resulted from temperature variation was in the magnitude of 10⁻⁵ (1/kΩ), far less than the precision 10⁻³ (1/kΩ) of the measuring apparatus.

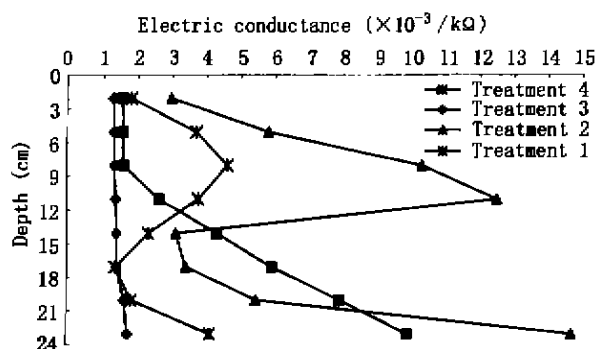


Fig.1. Comparison of soil water content profiles.

2.2 Vertical distribution of roots

The profiles of total lateral root length obtained by the root harvesting and total root biomass obtained by the soil coring were extremely spatially heterogeneous (Figs. 2,3).

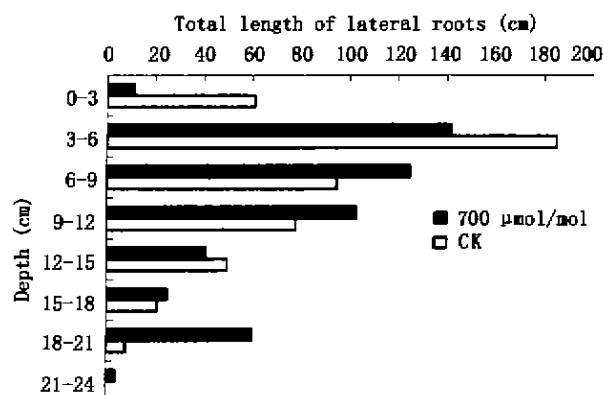


Fig.2. Vertical distribution of lateral roots.

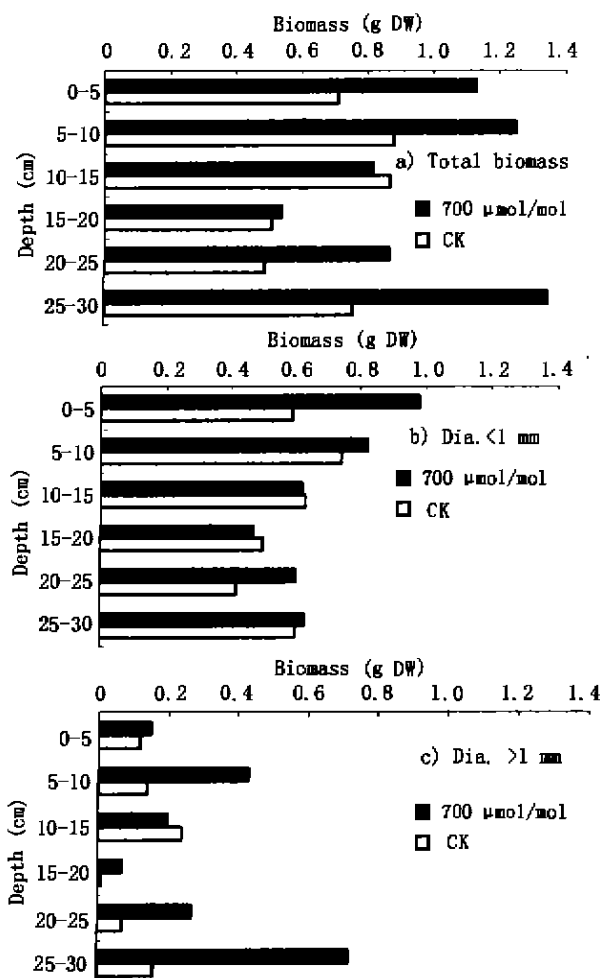


Fig.3. Vertical distribution of roots biomass. Dia., diameter.

Under the ambient CO₂ concentration, the majority of roots dominated by fine roots mainly distributed in the 0–15 cm soil layer, and the biomass of roots near the interface was very little. Under doubled CO₂, there were two peak zones of the root biomass, one was at 0–15 cm soil layer, and the other at 20–30 cm across the interface zone. Total root biomass increased more under doubled CO₂ no matter whether it was in the surface or interface zone, biomass increase on the surface was mainly attributed to the fine roots

of no more than 1 mm in diameter, in the interface area, it was attributed to those of more than 1 mm in diameter.

High spatial heterogeneity was one of the basic features of the distribution of tree roots. It depends on soil texture, water and nutrient availability^[12]. In our experimental sites, the heterogeneity of soil texture was great, especially across the interface area, because of the difference of soil nutrient and bulk density, the presence of soil soft spots and hollow chambers etc. Sternberg *et al.*^[8] used stable isotopes to trace the source of roots in the forest soil and pointed out that the root distribution was not homogeneous in either the horizontal or vertical direction in this kind of heterogeneous soil.

Fine root generally concentrates in the uppermost 10–20 cm of mineral soil. The fine root biomass largely distributed near the soil surface, this has been the subjects of many recent studies on seasonal patterns and carbon costs^[12]. Elevated atmospheric CO₂ usually increases the dimension of plants and their component parts^[13]. Our study showed that elevated CO₂ led to the increase of fine-root number or biomass in the interface and the surface zone, indicating that root activity was enhanced and the microenvironment of roots was improved. Therefore, the ability of material exchange between soil and roots, i.e. the water movement, increased.

The soil water profile was found to be similar to the root distribution profile, and the profile overlapped peak positions. It can be inferred that there was a robust positive correlation between soil water and root distribution. In the surface and the interface zones, the greater the root biomass, the stronger the water transport between soil and roots is. Elevated atmospheric CO₂ aggravated the stress of the soil water and made *Pinus sylvestris* var. *sylvestriformis* seedling roots develop toward deeper soil layer and grow much more roots in the interface zone. Abundant roots will enhance its activity on the surface and the interface area^[7], and luxuriant roots will enhance the ability of water transport between the soil and the seedling roots, for this reason, the soil water state will improve greatly at the surface and the interface areas. It is noteworthy that the root biomass and the soil water content were maximum and that the ability of water transport was the greatest in the 5 cm soil layer above the interface area. The coupling process of the underground roots distribution with soil water is more sensitive at the interface area, as supported by Rogers and Runion^[13] and Johnson *et al.*^[7]. Rogers and Runion^[13] pointed out that the dry weight, length and volume of taproots, lateral roots and fine roots tended to be higher for plants under elevated CO₂. Johnson *et al.*^[7] thought that the carbon at a given depth was more sensitive to the change of soil moisture.

The soil electrical conductance profile could be considered as an indication of the state of water transport between

soil and roots. This method can also provide a kind of inexpensive, non-destructive and relatively sensitive indices of the response of underground water transport to elevated CO₂.

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